

SOFT COMPUTING CAPSTONE

SC01

Intelligent Campus Climate and Energy Controller

Beginner Project Guide

All steps at a glance, with a fully spoon-fed Step 1.

Team size: 3-4 students

Step 1 target: 3-5 working days

M1: one week before mid-semester examination

M2: one week before end-semester examination

What are you building?

Build a small app that looks at room conditions and suggests how much cooling and fan speed are needed. It should also avoid wasting energy when the room is empty or electricity is expensive.

Who will use it?

A campus facility manager who wants a clear room-cooling recommendation.

One-sentence problem

Given room temperature, humidity, occupancy, and tariff, recommend cooling, fan level, and an energy action.

What M1 must show

A fuzzy controller for one room, compared with a simple temperature-rule controller.

What M2 must add

Use a Genetic Algorithm (GA) to improve the fuzzy settings, test difficult weather cases, and deploy the app.

Important

M1 must be a working Product V1: input -> processing -> output. Step 1 is the preparation needed before the main fuzzy, ANN, or GA algorithm. M2 must add a real technical improvement and a deployed Product V2.

Your eight-step project path

Step	Stage	What happens
1	M1	Choose one room and define its inputs and outputs.
2	M1	Create and validate climate and occupancy scenarios.
3	M1	Build a simple fixed-threshold controller.
4	M1	Build the fuzzy controller and explain its rules.
5	M1	Join everything in a working Product V1 screen.
6	M2	Use GA to tune membership functions or rules.
7	M2	Compare fixed, fuzzy, and GA-fuzzy results.
8	M2	Deploy Product V2 and finish all evidence.

Where this guide spends most of its time

The remaining pages explain Step 1 in very small, doable actions. Finish every checkbox before starting the main algorithm.

Do not begin fuzzy rules, neural-network training, or GA optimization just because one member is excited to code. First agree on the problem, fields, units, sample data, baseline, tests, and evidence.

What Step 1 must produce

- ☐ A GitHub repository that every member can open.
- ☐ A README.md that explains the project and how the team will work.
- ☐ A docs/STEP-1.md project contract.
- ☐ A data/README.md with field meanings, units, sources, and assumptions.
- ☐ The correct starter data files.
- ☐ Twenty clean starter rows or one small, complete fixture.
- ☐ Five named test cases, including an invalid or impossible case.
- ☐ A simple baseline written as numbered rules or pseudocode.
- ☐ A validation program that prints PASS for valid data and rejects bad data.
- ☐ A simple Product V1 screen sketch.
- ☐ At least one meaningful Git commit from every member.
- ☐ A three-minute faculty demonstration.

Suggested five-day plan

1	Day 1: create repository, add members, assign roles, and write the one-sentence problem.
2	Day 2: define every input/output field, unit, range, category, and missing-value rule.
3	Day 3: create the 20-row sample or small fixture; write the five test cases.
4	Day 4: write baseline rules, validation code, screen sketch, and make member commits.
5	Day 5: clone/run from the beginning, correct errors, and rehearse the faculty demo.

Create the working space

1. One member creates a new repository. Suggested name:

SC01-Intelligent-Campus-Climate-and-Energy-Controller-Team-Name

2. Add all members as collaborators. Every member accepts the invitation.

3. Add these folders and files:

```
README.md
requirements.txt
docs/STEP-1.md
data/README.md
data/sample_input.csv # or the project-specific starter files
src/validate_data.py
notebooks/
app/
results/
report/
```

4. Assign first responsibilities

Responsibility	Exact Step 1 job
Repository	Create folders, add members, and test clone/run steps.
Data	Write fields/units, make the sample, and record sources.
Baseline	Write simple rules and expected results for five cases.
Testing/UI	Write validation, draw the screen, and collect evidence.

For three students, combine Repository with Testing/UI. These are first responsibilities, not permanent silos. Everyone must understand the complete Step 1.

5. If using Git commands

```
git clone REPOSITORY-URL
cd REPOSITORY-NAME
git checkout -b your-name-step1
# create or edit your assigned files
git add .
git commit -m "Describe the real work you completed"
git push -u origin your-name-step1
```

If Git commands feel difficult, use GitHub Desktop. The required idea is the same: pull first, work on a small task, write a clear commit message, push, and let the team review.

Write docs/STEP-1.md

Copy these headings into **docs/STEP-1.md**. Then replace every instruction with your team's answer.

Step 1 Project Contract

Team and responsibilities

- Member name - first responsibility

One-sentence problem

Given room temperature, humidity, occupancy, and tariff, recommend cooling, fan level, and an energy action.

User of the product

A campus facility manager who wants a clear room-cooling recommendation.

Inputs and units

Write the field table from the next page.

Outputs and units

cooling_pct (0-100), fan_level (off/low/medium/high), energy_action (normal/save/avoid_peak)

Baseline method

Write the numbered baseline rules from this guide.

Soft Computing method for M1

A fuzzy controller for one room, compared with a simple temperature-rule controller.

Advanced method for M2

Use a Genetic Algorithm (GA) to improve the fuzzy settings, test difficult weather cases, and deploy the app.

Dataset/scenario sources

Add real URLs, licences, and assumptions.

Five mandatory test cases

Copy and complete the five cases in this guide.

Product V1 screen sketch

Insert a photo or exported image of the sketch.

Risks and assumptions

List what may be wrong, missing, simulated, or temporary.

Step 1 completion evidence

Add links to files, commits, and validation output.

Before saving

[] Replace generic words such as 'user' or 'data' with project-specific words.

[] Write units beside every number.

[] Mark temporary thresholds and assumptions as temporary.

[] Do not claim that simulated data is real data.

[] Do not paste a source URL without saying what information came from it.

Create the data dictionary

Field	Simple meaning	Type/unit	Starter rule
scenario_id	Unique row name	text	SC01-001, SC01-002, ...
temperature_c	Room temperature	number in C	18 to 45
humidity_pct	Relative humidity	percentage	20 to 100
occupancy_count	People in the room	whole number	0 to chosen room capacity
tariff_level	Current electricity price band	category	low, medium, high

For every field, the team must answer

- ☐ What does this field mean in one simple sentence?
- ☐ Is it a number, category, text identifier, input, target, or output?
- ☐ What is its unit?
- ☐ What values are allowed for the 20-row sample?
- ☐ What should happen when the value is missing?
- ☐ Which source or assumption justifies the final range?

Write the outputs separately

Required outputs: cooling_pct (0-100), fan_level (off/low/medium/high), energy_action (normal/save/avoid_peak)

Source and 10,000-record plan

Find cited temperature/humidity ranges, room-capacity assumptions, and tariff bands. Later generate at least 10,000 valid combinations from those ranges.

Step 1 uses only a small sample. The final 10,000+ records/scenarios are created in Step 2, after the small sample and validation are correct.

Create the first sample carefully

Do not begin with 10,000 rows. First make a small file that every member understands.

How to create the sample

1	Create the file or files listed on the repository page.
2	Type the field names exactly as written in the data dictionary.
3	Create five rows for the mandatory cases shown below.
4	Add five ordinary rows with different valid values.
5	Add five boundary rows close to minimum or maximum values.
6	Add five difficult rows that combine conditions.
7	Keep one wrong row in a separate test file. Do not mix it into valid data.
8	Save as CSV using comma-separated values and a header row.

Five mandatory starter cases

Name	Example input	What to check
SC01-001	24, 50, 20, low	Comfortable room; low cooling expected
SC01-002	36, 85, 45, medium	Hot and humid; strong cooling/fan expected
SC01-003	38, 60, 0, high	Empty room; energy-saving rule must apply
SC01-004	45, 70, 50, high	Heat-wave boundary case
SC01-005	21, 90, 15, low	Cool but humid; fan decision should be checked

After entering the rows

- ☐ Open the CSV once in a text editor and once in a spreadsheet.
- ☐ Check that no heading contains an accidental space.
- ☐ Check that units are not mixed inside a numeric column.
- ☐ Check that categories use one spelling only.
- ☐ Check that identifiers are unique.
- ☐ Ask a member who did not create the file to explain three rows.

Write the comparison method

The baseline is deliberately simple. It lets you prove later whether the Soft Computing method is better.

1	If occupancy_count is 0, set cooling_pct to 0 and fan_level to off.
2	Else if temperature_c is below 24, set cooling_pct to 10.
3	Else if temperature_c is 24 to 30, set cooling_pct to 50.
4	Else set cooling_pct to 90.
5	If humidity_pct is above 75, increase fan_level by one band.
6	If tariff_level is high, return energy_action as avoid_peak.

What to save

- [] Put the numbered rules in docs/STEP-1.md.
- [] Write the same rules as pseudocode in docs/baseline-pseudocode.md.
- [] For each of the five mandatory cases, write the expected baseline behavior.
- [] Mark every temporary threshold clearly.
- [] Do not call this baseline fuzzy logic, ANN, or GA.

Pseudocode format

```

READ one valid input row
IF the row is invalid:
    SHOW a clear error
ELSE:
    APPLY the simple rules in order
    CALCULATE the baseline output
    PRINT the output and the rule that was used
SAVE the result for later comparison
  
```

Step 1 requires the baseline design and expected behavior. A complete baseline program is built in Step 3, but writing a tiny runnable version now is encouraged.

Make the computer check your data

Install pandas and run the starter validator. The first version checks file shape, headings, missing values, and simple data types.

```
python -m venv .venv
# Windows PowerShell
.venv\Scripts\Activate.ps1
pip install pandas
pip freeze > requirements.txt
python src/validate_data.py
```

Starter content for src/validate_data.py

```
import pandas as pd

FILE = 'data/sample_input.csv'
REQUIRED = ['scenario_id', 'temperature_c', 'humidity_pct', 'occupancy_count', 'tariff_level']
NUMERIC = ['temperature_c', 'humidity_pct', 'occupancy_count']

df = pd.read_csv(FILE)
print('shape:', df.shape)
print('columns:', list(df.columns))
print('missing values:', df.isna().sum().to_dict())

missing_columns = [c for c in REQUIRED if c not in df.columns]
assert not missing_columns, f'Missing columns: {missing_columns}'
assert not df[REQUIRED].isna().any().any(), 'Missing value found'
for column in NUMERIC:
    assert pd.api.types.is_numeric_dtype(df[column]), f'{column} must be numeric'

print('STEP 1 DATA CHECK PASSED')
```

Add project-specific checks

- ☐ Every percentage must stay inside 0 to 100.
- ☐ Counts, time, distance, power, money, or rainfall cannot be negative.
- ☐ Every category must match the documented spelling.
- ☐ Every identifier must be unique.
- ☐ The known invalid/impossible case must fail with a clear message.
- ☐ The five valid mandatory cases must pass.

Take a screenshot of one PASS result and one expected FAIL result. Save them in results/step1/.

Draw the screen before coding it

Use paper, PowerPoint, Figma, Canva, or a simple drawing tool. The sketch is not the final UI.

Your screen must show

- [] Input area: Temperature, humidity, occupancy, tariff.
- [] One clear Run, Recommend, Predict, or Optimize button.
- [] Output area: Cooling %, fan level, energy action, baseline result, explanation.
- [] A place for validation errors.
- [] A place for one explanation or chart.
- [] A visible difference between baseline output and future Soft Computing output.

Simple wireframe to copy

```
+-----+
| PROJECT TITLE                |
+-----+
| INPUTS      | RESULTS      |
| [field] [value] | Baseline: ... |
| [field] [value] | Soft Computing: ... |
| [field] [value] | Explanation: ... |
| [ RUN / RECOMMEND ] | [ chart or visual area ] |
+-----+
| Validation / error message    |
+-----+
```

Save the sketch

1	Export or photograph the sketch clearly.
2	Save it as docs/product-v1-sketch.png.
3	Insert it in docs/STEP-1.md.
4	Ask one person outside the team to explain what they think the screen does.
5	Correct any label that the person does not understand.

Make small, honest commits

Each commit should contain one understandable improvement. Every member must make at least one commit.

Recommended commit sequence

1	Create repository structure for SC01
2	Add team roles and Step 1 project contract
3	Define input fields, units, and validation rules
4	Define climate input schema and five test cases
5	Add five mandatory test cases
6	Add sample-data validation script
7	Add Product V1 screen sketch
8	Add Step 1 evidence and run instructions

Before every push

- ☐ Pull the latest team version.
- ☐ Run the validation program.
- ☐ Check git status and add only intended files.
- ☐ Write a message that says what changed.
- ☐ Push your branch and ask another member to review.

Do not do this

- ☐ Do not use messages such as update, changes, final, or done.
- ☐ Do not upload the complete project in one last commit.
- ☐ Do not commit .venv, passwords, private datasets, or personal information.
- ☐ Do not overwrite another member's work without discussing it.

Exactly what to show faculty

Open the repository and demonstrate the following in order.

1	Show the repository URL and collaborator list.
2	Open README.md and state the one-sentence problem.
3	Open docs/STEP-1.md and explain the user, inputs, outputs, M1, and M2.
4	Open data/README.md and show the field meanings, units, source links, and assumptions.
5	Open the sample data and point out one normal, one boundary, and one difficult row.
6	Run <code>python src/validate_data.py</code> and show PASS.
7	Run or show the deliberately invalid case and explain the expected error.
8	Explain the baseline rules and expected result for one test case.
9	Show the Product V1 screen sketch.
10	Show Git history and one meaningful commit from every member.

Final Step 1 checklist

- ☐ Every required file exists and opens.
- ☐ The problem boundary is small and clear.
- ☐ Inputs and outputs have names, meanings, units, and rules.
- ☐ The starter data is understandable and valid.
- ☐ Five mandatory cases are written with expected behavior.
- ☐ The simple baseline is explainable.
- ☐ Validation shows both PASS and expected FAIL evidence.
- ☐ The screen sketch includes input, result, explanation, and errors.
- ☐ Every member has committed and can explain the work.

After faculty approval, begin Step 2: create the reproducible 10,000+ record/scenario pipeline. Do not change field meanings or units silently; update the contract and commit the change.

What comes after Step 1

Step	Stage	What happens
1	M1	Choose one room and define its inputs and outputs.
2	M1	Create and validate climate and occupancy scenarios.
3	M1	Build a simple fixed-threshold controller.
4	M1	Build the fuzzy controller and explain its rules.
5	M1	Join everything in a working Product V1 screen.
6	M2	Use GA to tune membership functions or rules.
7	M2	Compare fixed, fuzzy, and GA-fuzzy results.
8	M2	Deploy Product V2 and finish all evidence.

M1 finish line

- ☐ 10,000+ cited records or documented scenarios
- ☐ Working baseline
- ☐ Working core Soft Computing method
- ☐ Runnable Product V1
- ☐ At least two useful result visuals
- ☐ Interim report and M1 demonstration

M2 finish line

- ☐ Advanced or hybrid method
- ☐ Parameter experiments and difficult-case testing
- ☐ Baseline vs M1 vs M2 comparison
- ☐ Public deployment and usable interface
- ☐ Final report, approximately five-minute video, presentation, and viva preparation

A strong capstone is one that another person can open, run, understand, test, and compare.